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builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/switch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-unknown-elf-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/dua_ansari/xv6-labs-2025'
== Test answers-traps.txt ==
answers-traps.txt: OK
== Test backtrace test ==
$ make qemu-gdb
backtrace test: OK (4.7s)
== Test running alarmtest ==
$ make qemu-gdb
(7.2s)
== Test alarmtest: test0 ==
alarmtest: test0: OK
== Test alarmtest: test1 ==
alarmtest: test1: OK
== Test alarmtest: test2 ==
alarmtest: test2: OK
== Test alarmtest: test3 ==
alarmtest: test3: OK
== Test usertests ==
$ make qemu-gdb
usertests: OK (172.3s)
== Test time ==
time: OK
Score: 95/95

```

Backtrace output

```

error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/DuaAliAnsari/xv6-labs-2025.git'
dua_ansari@DESKTOP-4DF8PF8:~/xv6-labs-2025$ git push origin traps
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 4 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 1.30 KiB | 443.00 KiB/s, done.
Total 9 (delta 7), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (7/7), completed with 7 local objects.
To https://github.com/DuaAliAnsari/xv6-labs-2025.git
dalf383..024f34a traps -> traps
dua_ansari@DESKTOP-4DF8PF8:~/xv6-labs-2025$ nano answers-traps.txt
dua_ansari@DESKTOP-4DF8PF8:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_btttest user/_alarmtest user/_call user/_btttest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1073 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ btttest
backtrace:
0x0000000000001f20
0x0000000000001d84
0x0000000000001aac
$ |

```

Alarmtest output

```
dua_ansari@DESKTOP-4DF8P  X  +  v
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ alarmtest
test0 start
.....alarm!
test0 passed
test1 start
..alarm!
.alarm!
.alarm!
.alarm!
.alarm!
.alarm!
.alarm!
..alarm!
.alarm!
.alarm!
.alarm!
test1 passed
test2 start
.....alarm!
test2 passed
test3 start
test3 passed
$ |
```